

UNSW Sydney, MATH2601 Practice Problems, T2 2025

[K] - key question, [H] - hard question, [L] - long question

Revision: Linear Equations and Matrices

1. Find the solutions to the following systems of equations. If possible give a geometric interpretation of the solution.

$$\begin{array}{lcl} \text{(a)} & 3x_1 + 2x_2 + x_3 = 10 & \text{(b)} \quad 3x_1 + 2x_2 + x_3 + x_4 = 10 \\ & 4x_2 + 2x_3 = 8 & 4x_2 + 2x_3 - 4x_4 = 8 \\ & -7x_3 = 14 & -7x_3 + 14x_4 = 14 \end{array}$$

2. For which values of λ do the equations

$$\begin{array}{rcl} x + 2y + \lambda z & = & 1 \\ -x + \lambda y - z & = & 0 \\ \lambda x - 4y + \lambda z & = & -1 \end{array}$$

have (a) no solution, (b) an infinite number of solutions. (c) a unique solution.

- [K]3. Given the matrices

$$A = \begin{pmatrix} -1 & -3 & 4 \\ 3 & 2 & -2 \\ 1 & -2 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & 1 \\ 3 & 4 \\ -1 & 5 \end{pmatrix}, \quad C = \begin{pmatrix} -3 & 2 \\ 1 & -4 \\ 6 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 3 & 1 \\ 1 & -2 & -3 \end{pmatrix}.$$

Find the following matrices if they exist, or explain why they do not exist.

$$\begin{array}{lllll} \text{(a)} & AB & \text{(b)} & BA & \text{(c)} & CD & \text{(d)} & DC & \text{(e)} & A^2 \\ \text{(f)} & C^2 & \text{(g)} & A^{-1} & \text{(h)} & B^{-1} & \text{(i)} & (CD)^{-1} & \text{(j)} & (DC)^{-1} \end{array}$$

4. Suppose A and B are matrices such that both AB and BA are defined.

- Show that AB and BA are both square matrices.
- If $AB = BA$, show that A and B are both square and of the same size.
- If A and B are square matrices such that $AB = BA$, show that $(A - B)(A + B) = A^2 - B^2$.
- Find two 2×2 matrices A, B for which $(A - B)(A + B) \neq A^2 - B^2$.
- Prove that $(A + B)^2 = A^2 + B^2 + 2AB$ if and only if $AB = BA$.

5. Let

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \\ -1 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 2 \\ 3 & -2 \\ -2 & 4 \end{pmatrix}.$$

Show that $AB = AC$ and deduce that matrices cannot in general be cancelled from products.

- [K]6. Let $\mathbf{a} = (1, 3, -2)^T$ and $\mathbf{b} = (0, 4, 2)^T$. Evaluate all of the following expressions that make sense and find those which are equal:

$$\mathbf{ab}, \quad \mathbf{a}^T \mathbf{b}, \quad \mathbf{ab}^T, \quad \mathbf{a}^T \mathbf{b}^T, \quad \mathbf{b}^T \mathbf{a}, \quad \mathbf{ba}^T.$$

7. Show by constructing an example that, in general, $A^T A \neq A A^T$, even if A is square.

- [K]8. Evaluate the determinants of the following 2×2 matrices and hence determine whether or not they are invertible.

$$(a) \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}, \quad (b) \begin{pmatrix} -4 & 7 \\ 3 & -5 \end{pmatrix}, \quad (c) \begin{pmatrix} 5 & 2 \\ 10 & 4 \end{pmatrix}, \quad (d) \begin{pmatrix} 8 & 9 \\ 3 & 4 \end{pmatrix}, \quad (e) \begin{pmatrix} 11 & 13 \\ 12 & 14 \end{pmatrix}.$$

Write down the inverses where they exist.

- [K]9. Evaluate the determinants for the following matrices by reducing to row echelon form.

$$(a) \begin{pmatrix} -1 & 1 & 2 \\ 2 & 4 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \quad (b) \begin{pmatrix} 1 & -2 & 4 \\ 3 & 1 & -2 \\ 1 & 5 & -10 \end{pmatrix}, \quad (c) \begin{pmatrix} 1 & 0 & 4 \\ 3 & 1 & -2 \\ 1 & 5 & -10 \end{pmatrix}.$$

Find the inverses where they exist.

1 Groups and Fields

- [K]10. Let n be an integer larger than 1 and write $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$. For $x, y \in \mathbb{Z}_n$, let $x * y$ be the (least non-negative) remainder when xy is divided by n . (That is, $*$ is multiplication modulo n .)

- (a) Prove that $*$ is a well-defined binary operation on $\mathbb{Z}_n$. Deduce from standard properties of arithmetic in $\mathbb{Z}$ that $*$ is commutative and associative.
- (b) Let $p \in \mathbb{Z}$ be prime. Show that $\mathbb{U}_p = \mathbb{Z}_p \setminus \{0\}$ is a group under $*$. (Bezout's identity will help: if x and y are relatively prime then there exist $a, b \in \mathbb{Z}$ such that $ax + by = 1$.)

11. Let $(G, *)$ be a group with identity e . Prove carefully, quoting the axioms as you use them, that each element of G has only one inverse, and that for each $a \in G$, we have $(a^{-1})^{-1} = a$.

12. Suppose that G is a group and that $a^2 = e$ for all a in G . Show that G is Abelian.

- [K]13. Show that if m_1, m_2 are integers and $m_1 \mid m_2$, then $m_2\mathbb{Z}$ is a subgroup of $m_1\mathbb{Z}$. (The operation on both groups is addition.)

14. (a) Show that the set of matrices

$$G = \left\{ \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a, b \in \mathbb{R}, a \neq 0 \right\}$$

is a subgroup of $\text{GL}(2, \mathbb{R})$, the group of 2×2 invertible matrices with entries in $\mathbb{R}$ under matrix multiplication.

- (b) Show that the set of matrices

$$H = \left\{ \begin{pmatrix} a & a \\ a & a \end{pmatrix} : a \in \mathbb{R}, a \neq 0 \right\}$$

is a group under matrix multiplication. Why is it *not* a subgroup of $\text{GL}(2, \mathbb{R})$?

- 15.** Recall that the *order* of a group G , denoted $|G|$, is the number of elements in the group. Let x be an element of G .

- (a) If $x^i = x^j$ with $i \neq j$ show that there is a least positive integer n such that $x^n = e$. If n is the smallest power of x such that $x^n = e$ then n is the *order* of x .
- (b) If G is a group of finite order and x is in G then show that the order of x is finite.
- (c) Show that if $\phi : G \rightarrow H$ is an isomorphism between G and another group H , then the order of every g in G is the same as the order of $\phi(g)$ in H .

- [K]16.** Let H be the smallest subgroup of $\text{GL}(2, \mathbb{R})$ containing both

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Show that H has eight elements.

(We call H the group *generated* by A and B and write $H = \langle A, B \rangle$.)

- 17.** Let Q be the subgroup of $\text{GL}(2, \mathbb{C})$ generated by $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$.

Show that Q is a group with 8 elements: it is a linear representation on $\mathbb{C}^2$ of the *quaternion group*.

- 18.** Using **Q15(c)**, or otherwise, show that H of **Q16** and Q from **Q17** are not isomorphic, and neither are isomorphic to C_8 , the cyclic group of order 8.
- 19.** (a) Let G be the group of symmetries of a cube. Show that $|G| = 24$. Describe the elements of G and classify them by their order. (See **Q15** for the definition of the order of an element.)
- (b) Repeat (a) for the group of symmetries of the dodecahedron. In this case the order of the group is 60.

- 20.** For any field $\mathbb{F}$ and $a, b, c \in \mathbb{F}$, prove the following:

- (a) $a0 = 0$,
- (b) $a(-b) = -(ab)$,
- (c) $a(b - c) = ab - ac$.

[K]21. Which of the following are fields? For the ones which are not fields, say which axiom fails.

- (a) $M_{2,2}(\mathbb{R})$ under the usual addition and multiplication of matrices.
- (b) $G = \{x + iy \in \mathbb{C} : x, y \in \mathbb{Q}\}$ under the usual addition and multiplication in $\mathbb{C}$.
- (c) $S = \{x + y\sqrt[3]{2} \in \mathbb{R} : x, y \in \mathbb{Q}\}$, with operations from $\mathbb{R}$.
- (d) $\mathbb{Z}_{109}$. (e) $\mathbb{Z}_{110}$.

22. For each of the following, say whether the statement is true or false and give a brief reason.

- (a) The function $\phi : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ with $\phi(n) = 2n$ is a group homomorphism.
- (b) The function $\phi : (\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$ with $\phi(x) = 2x$ is a group homomorphism.
- (c) The function $\phi : (\mathbb{R}, +) \rightarrow (\mathbb{R}, +)$ with $\phi(x) = 2x - 1$ is a group homomorphism.
- (d) The function $\phi : (\mathbb{R}^+, \times) \rightarrow (\mathbb{R}, +)$ with $\phi(x) = \log x$ is a group homomorphism.

23. For each function in **Q22** which is a homomorphism, which are isomorphisms?

24. Let $\phi : G \rightarrow H$ be a group homomorphism. Prove that:

- (a) $\phi(a^{-1}) = (\phi(a))^{-1}$ for all $a \in G$,
- (b) $\text{im}(\phi) \leq H$,
- (c) if ϕ is an isomorphism then so is ϕ^{-1} .

2 Vector Spaces

[K]25. For each of the following, say whether the statement is true or false and give a brief reason.

- (a) The set $\{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ is a vector space.
- (b) The set $\{(x, y, z) \in \mathbb{R}^3 : xyz = 0\}$ is a vector space.
- (c) The set $\{(x, y, z) \in \mathbb{R}^3 : \sin(x + y + z) = 0\}$ is a vector space.
- (d) The set $\{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^4\}$ is a vector space, where A is a fixed 3×4 real matrix.

26. Let V be the set of all sequences $\{\alpha_n\} = \{\alpha_0, \alpha_1, \dots, \alpha_n, \dots\}$ of elements of $\mathbb{R}$. Define addition of sequences by: $\{\alpha_n\} + \{\beta_n\} = \{\alpha_n + \beta_n\}$ and scalar multiplication by $\lambda\{\alpha_n\} = \{\lambda\alpha_n\}$. Prove that V is a vector space over $\mathbb{R}$.

27. Let $V = C([-1, 1])$, the vector space of all continuous real valued functions defined on $[-1, 1]$. Which of the following are subspaces of V ? Justify your answers.

- (a) $\{f \in V : f(t) = f(-t) \text{ for all } t\}$, (b) $\{f \in V : f(t) \geq 0 \text{ for all } t\}$,
- (c) $\{f \in V : \int_{-1}^1 f(t) dt = 0\}$. (d) $\{f \in V : \int_{-1}^1 f(t) dt = 1\}$.

[K]28. Are the following sets subspaces of $\mathbb{R}^3$? Give reasons for your answers:

- (a) $S_1 = \{\mathbf{x} : x_1^2 = x_2^3\}$;
- (b) $S_2 = \{\mathbf{x} : 2x_1 - 3x_2 + 3x_3 = 13\}$;
- (c) $S_3 = \{\mathbf{x} : 2x_1 - 3x_2 + 3x_3 = 0\}$;
- (d) $S_4 = \{\mathbf{x} : \mathbf{x} = t_1 \mathbf{u}_1 + t_2 \mathbf{u}_2, t_1, t_2 \in \mathbb{R}\}$ for some fixed vectors $\mathbf{u}_1, \mathbf{u}_2$ in $\mathbb{R}^3$.

Find a basis for those that are subspaces.

[K]29. Determine which of the following subsets of $\mathcal{P}_3(\mathbb{R})$ are subspaces.

- (a) $\{p \in \mathcal{P}_3(\mathbb{R}) : p''(x) = 0 \text{ for all } x \in \mathbb{R}\}$,
- (b) $\{p \in \mathcal{P}_3(\mathbb{R}) : p'(x) + x + 1 = 0 \text{ for all } x \in \mathbb{R}\}$.

30. Prove that the following are real vector spaces:

- (a) The set $\{\{\alpha_n\} : \alpha_{n+1} - \alpha_n = \alpha_{n+2} - \alpha_{n+1}, n \geq 0, \alpha_n \in \mathbb{R}\}$ of all *arithmetic progressions*.
- (b) The set $\{\{\alpha_n\} : \alpha_{n+2} = \alpha_{n+1} + \alpha_n, n \geq 0, \alpha_n \in \mathbb{R}\}$ of all *Fibonacci sequences*.
- (c) The set of all *convergent sequences*, defined as

$$\{\{\alpha_n\} : \exists \alpha \text{ such that } \forall \epsilon > 0 \exists N \in \mathbb{Z} \text{ such that } n > N \Rightarrow |\alpha_n - \alpha| < \epsilon\}.$$

[K]31. In each case, determine whether the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly independent. If it is not, express one vector as a linear combination of the other two.

- (a) $\mathbf{v}_1 = (1, 2, 3)^T, \mathbf{v}_2 = (2, 3, 4)^T, \mathbf{v}_3 = (3, 4, 5)^T$ in $\mathbb{R}^3$.
- (b) $\mathbf{v}_1 = (1, 2 + i, i)^T, \mathbf{v}_2 = (i, 1 - i, 1)^T, \mathbf{v}_3 = (5, 4 + i, i)^T$ in $\mathbb{C}^3$.

[K]32. Let V be the real vector space of continuous functions $f : [0, 2] \rightarrow \mathbb{R}^2$, and consider the elements f, g of V defined by $f(x) = (x, 1)$ and $g(x) = (4, x)$. Which of the following sets are linearly independent?

- (a) The subset $\{f, g\}$ of V .
- (b) For fixed $x \in [0, 2]$, the subset $\{f(x), g(x)\}$ of $\mathbb{R}^2$. (*Hint*: The answer depends on the value of x .)
- (c) The subset $\{f - g, f + g\}$ of V .
- (d) The subset $\{f, g, f + g\}$ of V .

33. Decide whether the following subsets of $\mathcal{C}([0, 2\pi])$ are linearly independent, and justify your answers:

- (a) $\{\cos x, \sin x\}$;
- (b) $\{\cos x, \sin x, \sin(x + \pi/4)\}$;
- (c) $\{\cos x, \sin x, \cos 2x\}$;
- (d) $\{\sin x, \sin 2x, \sin 3x\}$.

- 34.** Let V be the vector space of all twice-differentiable real-valued functions defined on the real line. For $f, g \in V$, we define the *Wronskian determinant* of f and g to be

$$W_{f,g}(t) = \det \begin{pmatrix} f(t) & g(t) \\ f'(t) & g'(t) \end{pmatrix}.$$

- (a) Prove that if there exists a t such that $W_{f,g}(t) \neq 0$ then $\{f, g\}$ is a linearly independent set in V .
- (b) Hence show that $\{\sin t, \cos t\}$ is a linearly independent set.
- (c) Show that $\{t^3, |t|^3\}$ has vanishing Wronskian but is linearly independent.

Note: Wronskian determinants play a significant role in the study of linear ordinary differential equations.

- 35.** Show that the set $\{t^2 + t^3, 1 + t + t^2 + t^3, t, 1 + t^2 + 2t^3\}$ is a linearly independent set in $\mathcal{P}_3(\mathbb{R})$. Hence prove that this set forms a basis for $\mathcal{P}_3(\mathbb{R})$ and find the coordinates of $1 + 2t + 3t^2 + 4t^3$ with respect to this basis.
- 36.** Show that $(1, 2)$ and $(3, 1)$ are linearly dependent in $\mathbb{Z}_5^2$. For which primes p are these vectors linearly independent in $\mathbb{Z}_p^2$?
- 37.** Show that $\mathcal{B} = \{(1, 2, 2)^T, (2, 1, 0)^T, (0, 0, 2)^T\}$ is a basis for $\mathbb{Z}_5^3$. Find the coordinates of $(4, 0, 0)^T$ with respect to this basis. For which primes p is $\mathcal{B}$ a basis for $\mathbb{Z}_p^3$?

- [K]38.** Let V be the vector space of all symmetric 2×2 matrices A with real entries, that is, A satisfies $A^T = A$. Show that

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 & -2 \\ -2 & 1 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \begin{pmatrix} 4 & -1 \\ -1 & -5 \end{pmatrix} \right\}$$

is a basis for V and find the coordinates of $\begin{pmatrix} 4 & -11 \\ -11 & -7 \end{pmatrix}$ with respect to this basis.

- 39.** Prove that in $\mathbb{F}^p$, if $P \in \text{GL}(p, \mathbb{F})$ is an invertible matrix and $\{\mathbf{v}_i\}$ is linearly independent, then the set $\{P\mathbf{v}_i\}$ is also linearly independent.

- [K]40.** Let $V = M_{2,2}(\mathbb{R})$ and let

$$W_1 = \left\{ \begin{pmatrix} x & -x \\ y & z \end{pmatrix} : x, y, z \in \mathbb{R} \right\} \text{ and } W_2 = \left\{ \begin{pmatrix} a & b \\ -a & c \end{pmatrix} : a, b, c \in \mathbb{R} \right\}.$$

Show that W_1 and W_2 are subspaces of V and find a basis for $W_1 + W_2$. Is this sum direct?

- 41.** (a) The set of all polynomials p of the form $p(t) = a_0 + a_1t + a_2t^2$, where $a_0, a_1, a_2 \in \mathbb{C}$, can be viewed either as a vector space over $\mathbb{C}$ which we will call V , or as a vector space over $\mathbb{R}$ which we will call W . Exhibit a basis for V and a basis for W .
- (b) Let L be the subset of such polynomials p which satisfy $p(1) \in \mathbb{R}$. Is L a subspace of W ? Is L a subspace of V ?

- (c) Let $X = \{p \in W : p(t) = a_0 + a_2 t^2, a_0 \in \mathbb{C}, a_2 \in \mathbb{R}\}$ and $Y = \{p \in W : p(t) = a_0 + a_1 t, a_0 + a_1 \in \mathbb{R}\}$. Find bases and the dimensions of X , Y , $X \cap Y$ and $X + Y$ in W .

[K]42. Suppose that W and X are subspaces of $\mathbb{R}^8$.

- (a) Show that if $\dim W = 3$, $\dim X = 5$ and $W + X = \mathbb{R}^8$, then $W \cap X = \{\mathbf{0}\}$.
 (b) Show that if $\dim W = \dim X = 5$, then $W \cap X$ has a linearly independent set with at least 2 elements.
 (c) Find an example of W and X with $\dim W = \dim X = 5$ and $\dim(W \cap X) = 3$.

3 Linear Maps

43. Let $T : V \rightarrow W$ and $S : W \rightarrow X$ be linear maps between vector spaces. Prove that $S \circ T : V \rightarrow X$ is also linear.

[K]44. For each of the following functions, either show that the function is linear and find its matrix with respect to the standard bases or prove that it is not linear.

- (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2a \\ a - b \end{pmatrix}$.
 (b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a + 1 \\ b \end{pmatrix}$.
 (c) $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$, $T(a + bt + ct^2) = (a + c) - (c + b)t + (a + b + c)t^2$.
 (d) $T : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$, $T(a + bt + ct^2) = a + b(t + 1) + c(t + 1)^2$.

- [K]45. (a) Given that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear and $T(3, 2)^T = (4, 6)^T$ and $T(2, 3)^T = (1, -1)^T$, find $T(4, 3)^T$.
 (b) Given that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear and $T(1, 2)^T = (1, -1)^T$ and $T(2, 3)^T = (3, 0)^T$, find a formula for $T(\mathbf{x})$.

46. If the matrix product AB exists, show that:

- (a) $\text{rank}(AB) \leq \text{rank}(A)$;
 (b) $\text{nullity}(AB) \geq \text{nullity}(B)$ and hence $\text{rank}(AB) \leq \text{rank}(B)$;
 (c) Deduce that $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.

[K]47. Find bases for the kernel (null space), the row space and the column space of

$$A = \begin{pmatrix} 1 & 3 & 1 & -2 & -3 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 3 & 8 & 1 & -7 & -8 \end{pmatrix}.$$

What are the nullity and rank of A ?

[K]48. Define $D : \mathcal{P}_2(\mathbb{R}) \rightarrow \mathcal{P}_2(\mathbb{R})$ by $D(p)(x) = p'(x)$. Show that D is linear and find the matrix of D with respect to the basis $\{1, 1+x, 1+x+x^2\}$ of $\mathcal{P}_2(\mathbb{R})$.

[K]49. Suppose that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear and has matrix $\begin{pmatrix} 4 & 9 \\ 1 & 1 \end{pmatrix}$ with respect to the standard basis of $\mathbb{R}^2$. What is the matrix of T with respect to the basis $B = \left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -3 \\ 2 \end{pmatrix} \right\}$?

50. Suppose that $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation such that $S \neq \mathbf{0}$ and $S \neq I$ and $S^2 = S$. Prove that there is a basis for $\mathbb{R}^2$ with respect to which the matrix A of S is

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

51. The vector space V has bases $B_1 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $B'_1 = \{\mathbf{v}'_1, \mathbf{v}'_2, \mathbf{v}'_3\}$ and W has bases $B_2 = \{\mathbf{w}_1, \mathbf{w}_2\}$ and $B'_2 = \{\mathbf{w}'_1, \mathbf{w}'_2\}$. The map $T : V \rightarrow W$ is linear and satisfies

$$T(\mathbf{v}_1) = \mathbf{w}'_1 + 3\mathbf{w}'_2, \quad T(\mathbf{v}_2) = 2\mathbf{w}'_1 + \mathbf{w}'_2, \quad T(\mathbf{v}_3) = 3\mathbf{w}'_1 + 2\mathbf{w}'_2.$$

Also, suppose that $\mathbf{v}'_1 = 2\mathbf{v}_1 + \mathbf{v}_2 + 3\mathbf{v}_3$, $\mathbf{v}'_2 = \mathbf{v}_1 + 5\mathbf{v}_2 + 6\mathbf{v}_3$, $\mathbf{v}'_3 = \mathbf{v}_1 + 3\mathbf{v}_2 + 3\mathbf{v}_3$ and $\mathbf{w}'_1 = 3\mathbf{w}_1 + \mathbf{w}_2$, $\mathbf{w}'_2 = 4\mathbf{w}_1 + \mathbf{w}_2$.

(a) Find the matrix A of T with respect to B_1 and B_2 .

(b) Find the matrix A' of T with respect to B'_1 and B'_2 .

[K]52. Let

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \quad \text{and} \quad Q = \begin{pmatrix} -1 & -2 & 2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{pmatrix}.$$

Define subspaces $V = \text{span}\{\mathbf{v}_1\}$ and $W = \text{span}\{\mathbf{v}_2, \mathbf{v}_3\}$ of $\mathbb{R}^3$.

(a) Show that $\mathbb{R}^3 = V \oplus W$.

(b) Show that V and W are invariant under the map T with $T(\mathbf{x}) = Q\mathbf{x}$.

(c) Find the matrix of T in part (b) with respect to the basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ of $\mathbb{R}^3$.

53. Show that the map $T : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by

$$T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}\right) = \begin{pmatrix} x_2 y_2 \\ -x_1 y_1 \\ x_1 y_1 - x_2 y_2 + x_2 y_1 \end{pmatrix}$$

is bilinear. Is T symmetric? Is it alternating? (Give proofs or counterexamples.)

4 Inner Product Spaces

[K]54. In $\mathbb{C}^2$, with the standard inner product, let $\mathbf{u} = (1 + 2i, 2)^T$, $\mathbf{v} = (2 + i, 1 - i)^T$ and $\mathbf{w} = (2 + 2i, -3 + i)^T$.

- (a) Find $\mathbf{u} \cdot \mathbf{v}$ and $\mathbf{v} \cdot \mathbf{u}$.
- (b) Find a unit vector parallel to $\mathbf{u}$.
- (c) Show that $\mathbf{u}$ is orthogonal to $\mathbf{w}$.

[K]55. Which of the following formulas define inner products on $\mathbb{R}^2$? For those which do not, say which axiom(s) fail(s).

- (a) $\langle \mathbf{v}, \mathbf{w} \rangle = v_1 w_1 + 2v_2 w_2$, (b) $\langle \mathbf{v}, \mathbf{w} \rangle = 2v_2 w_2$, (c) $\langle \mathbf{v}, \mathbf{w} \rangle = v_1 w_2 + 2v_2 w_1$.

[K]56. Which of the following formulas define inner products on $\mathbb{C}^2$? For those which do not, say which axiom(s) fail(s).

- (a) $\langle \mathbf{v}, \mathbf{w} \rangle = \bar{v}_1 w_1 + 2\bar{v}_2 w_2$, (b) $\langle \mathbf{v}, \mathbf{w} \rangle = \bar{v}_1 w_2 + 2i\bar{v}_2 w_1$.

57. Let $p(t) = a_0 + a_1 t + a_2 t^2$, $q(t) = b_0 + b_1 t + b_2 t^2 \in P_2(\mathbb{R})$. Which of the following define inner products on $P_2(\mathbb{R})$?

- (a) $\langle p, q \rangle = a_0 b_0 + a_1 b_1 + a_2 b_2$;
- (b) $\langle p, q \rangle = p(0)q(0) + p(1)q(1) + p(2)q(2)$;
- (c) $\langle p, q \rangle = p(0)q(0)$;
- (d) $\langle p, q \rangle = a_0 b_2 + a_1 b_1 + a_2 b_0$;

58. Show that the following is an inner product on $\mathbb{C}^2$.

$$\langle \mathbf{z}, \mathbf{w} \rangle = \bar{z}_1 w_1 + (1 + i)\bar{z}_1 w_2 + (1 - i)\bar{z}_2 w_1 + 3\bar{z}_2 w_2.$$

Find $\|(1 - 2i, 2 + 3i)^T\|$ with respect to the norm defined by this inner product.

59. Prove the following:

- (a) ([K]) $\langle \mathbf{x}, \mathbf{y} \rangle = \frac{1}{4}(\|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2)$ for real inner product spaces;
- (b) $\langle \mathbf{x}, \mathbf{y} \rangle = \frac{1}{4}(\|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2 - i\|\mathbf{x} + i\mathbf{y}\|^2 + i\|\mathbf{x} - i\mathbf{y}\|^2)$ for complex inner product spaces.

The formulae of (a) and (b) are known as the *polarization identities*.

[K]60. In each case, use the Gram-Schmidt process to find an orthonormal basis for the space spanned by the given vectors.

- (a) $\left\{ \begin{pmatrix} 5 \\ 12 \end{pmatrix}, \begin{pmatrix} -4 \\ 6 \end{pmatrix} \right\};$ (b) $\left\{ \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ -8 \end{pmatrix} \right\};$
- (c) $\left\{ \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\};$ (d) $\left\{ \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}, \begin{pmatrix} 1+i \\ 0 \\ 2 \end{pmatrix} \right\}.$

- 61.** Use Maple (or some other computer algebra system) to find polynomials $p_0(t), p_1(t), \dots, p_{10}(t)$ such that $p_i(t)$ has degree i and $\int_{-1}^1 p_i(t)p_j(t) dt = 0$ if $i \neq j$. Explain why each $p_i(t)$ with i odd is an odd function of t .

- 62.** Let V be an n -dimensional inner product space. If U and W are subspaces of V , prove that

$$(U + W)^\perp = U^\perp \cap W^\perp \quad \text{and} \quad (U \cap W)^\perp = U^\perp + W^\perp.$$

- [K]63.** (a) Calculate $W^\perp$ in $\mathbb{R}^3$ where $W = \text{span}\{(1, 1, 1)^T, (1, -1, 1)^T\}$.
 (b) Let V be the subspace of $\mathbb{R}^4$ spanned by $\{(1, 1, 1, -1)^T, (2, 1, 1, 1)^T, (1, 2, 0, 1)^T\}$. Find a basis for the orthogonal complement $V^\perp$ of V in $\mathbb{R}^4$.

- 64.** Let A be the matrix $\begin{pmatrix} 1 & 1 & -1 & -1 \\ 1 & 2 & -2 & 1 \end{pmatrix}$, let $W = \ker(A)$ and let $W^\perp$ be the orthogonal complement for W in $\mathbb{R}^4$.

- (a) Find orthonormal bases for W and $W^\perp$.
 (b) For $\mathbf{v} = (1, -1, 2, 3)^T$, find $\mathbf{w}_1 \in W$ and $\mathbf{w}_2 \in W^\perp$ such that $\mathbf{v} = \mathbf{w}_1 + \mathbf{w}_2$.

- [K]65.** In $\mathbb{C}^3$, find an orthonormal basis for $W^\perp$ if $W = \text{span}\{(1, i, 1)^T, (1 + i, 0, 2)^T\}$.

- [K]66.** Compute the adjoints of the following linear transformations.

- (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) := (2x + y, x)$ with the standard inner product.
 (b) $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ defined by $T(w, z) := (2w + iz, w)$ with the standard inner product.
 (c) $V = \mathbb{R}^2$ with inner product defined by $\langle v, w \rangle := 2v_1w_1 + v_2w_2$, and $T : V \rightarrow V$ defined by $T(v_1, v_2) := (2v_1 + v_2, v_1)$.

- 67.** Show that $A = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \\ 1 & -1 & 1 & -1 \end{pmatrix}$ is orthogonal. Such a matrix with all entries 1 and -1 (normalised) is called a *Hadamard matrix*. Can you find an 8×8 Hadamard matrix?

- 68.** Suppose that a matrix A is skew-adjoint (that is, $A^* = -A$).

- (a) Assuming that $I - A$ is invertible, show that $(I + A)(I - A)^{-1}$ is unitary.
 (b) (Harder) Prove that if $\mathbf{v} \in \mathbb{C}^n$ with $(I - A)\mathbf{v} = \mathbf{0}$ then $\mathbf{v} = \mathbf{0}$. (*Hint*: consider the inner product of $A\mathbf{v}$ with itself.)
 Why does this imply that $I - A$ is invertible?

- [K]69.** Show that $A = \begin{pmatrix} 5 & 9 \\ 12 & 11 \end{pmatrix} = \begin{pmatrix} 5/13 & 12/13 \\ 12/13 & -5/13 \end{pmatrix} \begin{pmatrix} 13 & 177/13 \\ 0 & 53/13 \end{pmatrix}$ is a QR-factorisation for the matrix A .

[K]70. Use the Gram-Schmidt method to find a QR-factorisation for each of the following matrices:

$$(a) \begin{pmatrix} 5 & -4 \\ 12 & 6 \end{pmatrix}; \quad (b) \begin{pmatrix} -2 & 1 \\ 1 & 4 \\ -2 & -8 \end{pmatrix}; \quad (c) \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 4 \end{pmatrix}; \quad (d) \begin{pmatrix} 1 & 2 & -1 \\ 1 & -1 & 2 \\ 1 & 2 & 2 \\ 1 & 1 & 1 \end{pmatrix}.$$

[K]71. Find the line $y = ax + b$ that best fits the three points $(1, 1)$, $(3, 2)$ and $(4, 6)$ in the least squares sense.

72. Let P, Q, R be the points $(1, 1)$, $(2, 1)$ and $(3, 1)$.

(a) Find the line $y = ax$ through the origin that best fits P, Q, R .

(b) Find the line $y = b + ax$ that best fits P, Q, R .

(c) Find the line $y = b + ax$ that best fits P, Q, R and the origin.

[K]73. For the points $(0, 0)$, $(1, 1)$, $(2, 3)$, $(-1, 5)$ find the least squares best-fit for:

(a) the line $y = a + bt$;

(b) the parabola $y = a + bt + ct^2$.

74. For the points

$$(1 + i, -i), \quad (i, -1), \quad (2i, i)$$

in $\mathbb{C}^2$ find the least squares best-fit line.

5 Determinants

75. Show that
$$\begin{vmatrix} a & b & c & d \\ -a & b & c & d \\ -a & -b & c & d \\ -a & -b & -c & d \end{vmatrix} = 8abcd.$$

[K]76. For what values of x is the determinant
$$\begin{vmatrix} x & a & b & c \\ a & x & b & c \\ a & b & x & c \\ a & b & c & x \end{vmatrix} = 0?$$

77. Factorize the determinant
$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}.$$

78. (a) Find $\det(tI - A)$ for $A = \begin{pmatrix} 0 & 0 & a_0 \\ 1 & 0 & a_1 \\ 0 & 1 & a_2 \end{pmatrix}.$

(b) Show that every polynomial whose leading coefficient is 1 (these are called *monic* polynomials) is equal to $\det(tI - A)$ for some matrix A .

[H]79. Recall that $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = (a-b)(b-c)(c-a).$

Show that if $a_i \in \mathbb{R}$, $1 \leq i \leq n$, then

$$\mathbb{V}_n = \begin{vmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} \end{vmatrix} = \prod_{1 \leq i < j \leq n} (a_j - a_i).$$

[Hint: One way is to show that the determinant is a polynomial in a_n of degree $n-1$ whose leading coefficient is $\mathbb{V}_{n-1}$. Then find all the zeros of this polynomial. A direct argument using the remainder theorem may be easier. You may also like to try the Maple `VandermondeMatrix` command.]

80. Using the previous question, or otherwise, show that if $(x_i, y_i) \in \mathbb{R}^2$ for $i = 1, 2, \dots, n$ with $x_i \neq x_j$ for $i \neq j$ then there is a unique polynomial p of degree $n-1$ with $p(x_i) = y_i$ for $i = 1, 2, \dots, n$.

Fiddly: Find an explicit formula for the polynomial p .

- [K]81. Show that a square matrix with integer coefficients has an inverse also with integer coefficients if and only if its determinant is ± 1 .

- [H]82. Show that the equation of the plane through three non-collinear points $\mathbf{a} = (a_1, a_2, a_3)$, $\mathbf{b} = (b_1, b_2, b_3)$ and $\mathbf{c} = (c_1, c_2, c_3)$ in $\mathbb{R}^3$ is

$$\begin{vmatrix} 1 & x_1 & x_2 & x_3 \\ 1 & a_1 & a_2 & a_3 \\ 1 & b_1 & b_2 & b_3 \\ 1 & c_1 & c_2 & c_3 \end{vmatrix} = 0.$$

Comment. It is actually quite difficult to give a complete answer to this question. Try this: (i) explain why the determinant, when expanded, has more or less the right form to be the equation of a plane, and why the points $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ satisfy this equation; (ii) identify any possible case in which the equation is not in fact that of a plane; (iii) prove that this case can only arise if the given points are collinear.

83. Consider the set of linear equations $A\mathbf{x} = \mathbf{b}$ where $A \in M_{p,p}(\mathbb{C})$ is invertible, so the solution is unique.

Define the matrix A_i as the same matrix as A but with its i th column replaced by $\mathbf{b}$.

Cramer's rule says that for the solution to $A\mathbf{x} = \mathbf{b}$, the element x_i of $\mathbf{x}$ is $\frac{\det(A_i)}{\det(A)}$.

- Check that Cramer's rule works for a 2×2 case by picking any invertible 2×2 matrix A and vector $\mathbf{x}$, calculating $A\mathbf{x}$ to get $\mathbf{b}$ and seeing what Cramer's rule gives you.
- Why is Cramer's rule not a practical way of solving a system of linear equations, except in the special cases of very sparse matrices? (A *sparse* matrix is one with a lot of zero entries.)
- Hard:* Prove Cramer's rule. (The adjugate formula for A^{-1} might help.)

6 Eigenvalues etc

- [K]84. Find all eigenvalues and eigenvectors of the following matrices. Are they diagonalizable over $\mathbb{R}$? over $\mathbb{C}$?

$$A = \begin{pmatrix} 8 & 6 \\ -2 & 1 \end{pmatrix}; \quad B = \begin{pmatrix} 2 & 0 & -1 \\ 0 & -3 & 2 \\ 0 & -5 & 3 \end{pmatrix}; \quad C = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

85. Find all eigenvalues and eigenvectors of the following matrices. Are they diagonalizable?

$$D = \begin{pmatrix} 1 & 2+i \\ 2+i & 1 \end{pmatrix}; \quad F = \begin{pmatrix} -2+i & -4 \\ 1 & 2+i \end{pmatrix}; \quad G = \begin{pmatrix} 1+i & 0 & 0 \\ 2-2i & 1-i & 0 \\ 2i & 0 & 1 \end{pmatrix}.$$

- [K]86. For each of the following matrices, use the given additional information to find all eigenvalues and eigenvectors *without calculating the characteristic polynomial*. Also write down the algebraic and geometric multiplicities of each eigenvalue.

(a) $A = \begin{pmatrix} 2 & -5 & -5 \\ -4 & 8 & 4 \\ 4 & -11 & -7 \end{pmatrix}$, given that 2 and -3 are eigenvalues;

(b) $B = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 1 & -2 \\ -3 & 4 & 6 \end{pmatrix}$, given that $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ are eigenvectors;

(c) $C = \begin{pmatrix} 6 & -2 & 1 \\ 2 & 1 & 2 \\ 3 & -6 & 8 \end{pmatrix}$, given that there is only one eigenvalue;

(d) $D = \begin{pmatrix} 4 & 1 & -1 & 2 \\ 2 & 5 & -2 & 4 \\ 4 & 4 & -1 & 8 \\ -1 & -1 & 1 & 1 \end{pmatrix}$, given that $\begin{pmatrix} -2 \\ 0 \\ 0 \\ 1 \end{pmatrix}$ is an eigenvector.

87. In each of the following cases explain, giving reasons, whether a matrix exists satisfying the given conditions:

- (a) A is a 4×4 matrix having eigenvalues $\lambda = 8$, with algebraic multiplicity 2 and geometric multiplicity 1; and $\lambda = -4$, with algebraic multiplicity 1; and $\lambda = 3$, with algebraic multiplicity 1;
- (b) B is a 7×7 matrix which has eigenvalues $\lambda = 3$ with algebraic multiplicity 2 and geometric multiplicity 2; and $\lambda = -1$ with algebraic multiplicity 6 and geometric multiplicity 4;
- (c) C is a 6×6 matrix having eigenvalues $\lambda = 1$, with algebraic multiplicity 3 and geometric multiplicity 2; and $\lambda = 2$, with algebraic multiplicity 2 and geometric multiplicity 2; and $\lambda = 3$, with algebraic multiplicity 1 and geometric multiplicity 2;

- (d) D is a *real* 2×2 matrix with spectrum $\{5 + 6i, 5 - 6i\}$;
 (e) E is a *real* 2×2 matrix with spectrum $\{1 + 2i, 3 + 4i\}$;
 (f) F is a *real* 5×5 matrix with eigenvalues $1 + 2i$ and $3 + 4i$, and possibly others.
- 88.** On $\mathcal{P}^3(\mathbb{R})$ define T by $T(p) = \tilde{p}$ where $\tilde{p}(t) = p(t+1)$ for $t \in \mathbb{R}$. Find all eigenvalues and eigenvectors of T .
- 89.** (a) Show that if diagonalisable matrices A and B have the same eigenvectors then $AB = BA$, that is A and B commute.
 (b) Suppose A has distinct eigenvalues. Show that any B that commutes with A has the same eigenvectors as A (and is thus diagonalisable).
 (c) Find an example of a diagonalisable A and a B that commutes with A but does not have the same eigenvectors as A .
- 90.** A *Markov matrix*, M , is a square matrix with all entries positive and such that the sum of the entries in each column is 1. Show that such a matrix has eigenvalue 1. (Markov matrices are used in population dynamics. A harder problem is to show that all other eigenvalues λ satisfy $|\lambda| \leq 1$.)
- [K]91.** (a) Prove that if $A = P^{-1}BP$ then $A^n = P^{-1}B^nP$ for any $n \in \mathbb{Z}$.
 (Note: this is a very useful result.)
 (b) Hence find $\begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}^{100}$, and (if it exists) $\lim_{n \rightarrow \infty} \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}^{-n}$.
 (c) Find all diagonalisable matrices M such that $M^2 = \begin{pmatrix} 3 & 1 \\ -2 & 0 \end{pmatrix}$.
- [K]92.** A linear map T on a finite dimensional inner product space V is called *skew-adjoint* if $T^* = -T$.
 (a) Prove that the eigenvalues of such a map are pure imaginary ($i\lambda$ for real λ).
 (b) Prove that if V is real then for any vector $\mathbf{v} \in V$, if T is skew-adjoint then $T(\mathbf{v})$ is orthogonal to $\mathbf{v}$.
- 93.** Show that every linear transformation T is the sum of a self-adjoint (Hermitian) transformation and a skew-adjoint transformation (see **Q92**).
Harder: What relationship (if any) holds between the eigenvalues of these three transformations when T is normal?
- 94.** Find an orthonormal basis of $\mathbb{R}^3$ consisting of eigenvectors for the matrix

$$A = \begin{pmatrix} 5 & 4 & 2 \\ 4 & 5 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

- 95.** Find the spectral form and orthonormal bases of eigenvectors for

$$A = \begin{pmatrix} 2 & -i \\ i & 2 \end{pmatrix}; \quad B = \begin{pmatrix} 2 & 2-i \\ 2+i & -2 \end{pmatrix}; \quad C = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}; \quad D = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

[K]96. For each of the following conic sections, find the principal axes, identify the curve and sketch it, showing the points closest to and (if appropriate) furthest from the origin.

$$(a) \ x^2 + 4xy + y^2 = 1; \quad (b) \ 3x^2 - 4xy + 3y^2 = 1; \quad (c) \ x^2 + 4xy + 4y^2 = 5.$$

97. For each of the following quadric surfaces, find the principal axes, identify the surface and find the points on the surface closest to the origin.

$$(a) \ 5x^2 + 3y^2 + 4z^2 + 4xz + 4yz = 1; \quad (b) \ x^2 + y^2 + 2z^2 - 2xz - 2yz = 1;$$

$$(c) \ 4z^2 + 2xy + 6xz + 6yz = 1; \quad (d) \ x^2 + y^2 + z^2 - 4xy - 4xz - 4yz = 3.$$

For each quadric surface find a rotation that puts the surface into canonical form and give the angles and the axis of the rotation.

[K]98. Identify (as rotations, reflections, etc.) the transformations on $\mathbb{R}^3$ with matrices

$$P = \frac{1}{9} \begin{pmatrix} -1 & 8 & -4 \\ 8 & -1 & -4 \\ -4 & -4 & -7 \end{pmatrix}; \quad Q = \frac{1}{2} \begin{pmatrix} 1 & 1 & -\sqrt{2} \\ -1 & -1 & -\sqrt{2} \\ -\sqrt{2} & \sqrt{2} & 0 \end{pmatrix}; \quad R = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 \\ -2 & 2 & 1 \\ 1 & 2 & -2 \end{pmatrix}.$$

Find the axis of rotation, the cosine of the angle of rotation, and (if appropriate) the plane of reflection.

99. Find the singular value decomposition, the reduced singular value decomposition, and the pseduoinverse of the matrices

$$A = \begin{pmatrix} -3 & -1 \\ 6 & 2 \end{pmatrix}; \quad B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{pmatrix}; \quad C = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -1 \end{pmatrix}.$$

7 Canonical Forms

[K]100. For each of the following, show that the matrix is not diagonalisable; explain why the only generalised eigenspace of the matrix is $\mathbb{C}^2$; find a basis consisting of a chain of generalised eigenvectors; and write the matrix in the form PJP^{-1} , where J is a Jordan form and P is invertible:

$$A = \begin{pmatrix} 10 & 1 \\ -9 & 4 \end{pmatrix}; \quad B = \begin{pmatrix} 5 & -3 \\ 12 & -7 \end{pmatrix}.$$

101. Show that each of the following matrices fails to be diagonalisable; explain why the only generalised eigenspace of the matrix is $\mathbb{C}^3$; find a basis consisting of one or more chains of generalised eigenvectors; and write the matrix in the form PJP^{-1} , where J is a Jordan form and P is invertible:

$$A = \begin{pmatrix} 5 & -1 & 3 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{pmatrix}; \quad B = \begin{pmatrix} 6 & 1 & 1 \\ -1 & 3 & -2 \\ 0 & 1 & 6 \end{pmatrix}; \quad C = \begin{pmatrix} 1 & -3 & 1 \\ 1 & 5 & -1 \\ 2 & 6 & 0 \end{pmatrix}.$$

[K]102. Express each of the following matrices as PJP^{-1} , where J is a matrix in Jordan form:

$$A = \begin{pmatrix} -5 & 6 & 6 \\ 2 & -8 & -5 \\ -2 & 5 & 2 \end{pmatrix} ; \quad B = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 5 & -8 \\ 1 & 4 & -7 \end{pmatrix} ; \quad C = \begin{pmatrix} -6 & 2 & 5 \\ -1 & -9 & -5 \\ 2 & 4 & 3 \end{pmatrix} .$$

[K]103. Given that A has eigenvalues 5 and 6 only, find a Jordan form J and change-of-basis matrix P for

$$A = \begin{pmatrix} 9 & 1 & -1 & -6 \\ 4 & 7 & -2 & -8 \\ -3 & -1 & 6 & 5 \\ 3 & 1 & -1 & 0 \end{pmatrix} .$$

[K]104. Are the matrices $\begin{pmatrix} 0 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and $\frac{1}{2} \begin{pmatrix} 2 & -5 & 1 \\ 0 & -1 & 3 \\ 0 & 1 & 1 \end{pmatrix}$ similar?

105. Determine which of the following matrices are similar to each other:

$$A = \begin{pmatrix} 1 & 1 & -1 \\ -8 & 7 & -4 \\ -4 & 2 & 1 \end{pmatrix} , \quad B = \begin{pmatrix} 1 & 2 & -1 \\ -2 & 5 & -1 \\ -1 & 1 & 3 \end{pmatrix} , \quad C = \begin{pmatrix} 5 & -2 & -2 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} , \quad D = \begin{pmatrix} 3 & 3 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} .$$

[K]106. Consider a 6×6 matrix having only one eigenvalue. How many “essentially different” Jordan forms are possible for such a matrix?

[K]107. Let A be a 12×12 matrix which has only one eigenvalue λ , and suppose that we determine $d_k = \dim \ker(A - \lambda I)^k$ for $k = 1, 2, \dots$. Which of the following sequences might we obtain? For those that are not possible, explain why not; for those that are, give a Jordan form of A and the minimal polynomial of A .

- (a) $d_k = 5, 9, 11, 13, 14, 14, \dots$;
- (b) $d_k = 5, 9, 11, 12, 12, 12, \dots$;
- (c) $d_k = 5, 11, 9, 12, 12, 12, \dots$;
- (d) $d_k = 4, 7, 9, 11, 12, 12, 12, \dots$;
- (e) $d_k = 4, 7, 11, 12, 12, 12, \dots$;
- (f) $d_k = 4, 9, 11, 12, 12, 12, \dots$;
- (g) $d_k = 3, 6, 9, 11, 12, 12, 12, \dots$.

[K]108. (a) A 9×9 matrix A has only one eigenvalue $\lambda = 3$. If

$$\text{nullity}(A - 3I) = 3 \quad \text{and} \quad \text{nullity}(A - 3I)^2 = 5 ,$$

find all possible Jordan forms of A . If A were given, what would you need to calculate in order to determine which of your possible Jordan forms is the correct one?

- (b) Repeat for a 14×14 matrix with only one eigenvalue $\lambda = -2$, if

$$\text{nullity}(A + 2I)^k = 4, 8, 11 \quad \text{for } k = 1, 2, 3$$

- (c) Repeat for a 12×12 matrix having

$$\text{nullity}(A - 7I)^k = 4, 6 \quad \text{for } k = 1, 2$$

$$\text{nullity}(A + 3I)^k = 2, 4, 5 \quad \text{for } k = 1, 2, 3.$$

- 109.** Show that a complex matrix is diagonalisable if and only if the minimal polynomial has distinct roots.

- [H]110.** What can be said about how the minimal polynomials of AB and BA are related? (*Hint:* in the singular case consider $tm(t)$ where $m(t)$ is the minimal polynomial of, say, AB .)

- [K]111.** Let $A = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix}$. Use the Cayley–Hamilton Theorem to find A^3, A^4, A^5 and A^{-1} as linear combinations of I, A and A^2 .

- 112.** Suppose that $N \in M_{n,n}(\mathbb{C})$ is *nilpotent* (that is, $N^k = \mathbf{0}$ for some integer $k > 0$). Show that $I + N$ is invertible, and find its inverse as a polynomial in N .

- [H]113.** Use the Jordan form to prove *Schur's Lemma*: if A is an $n \times n$ matrix over $\mathbb{C}$, then there exists an $n \times n$ unitary matrix Q and an upper triangular matrix U such that $A = QUQ^*$; moreover, the diagonal of U consists of the eigenvalues of A , repeated according to multiplicity.

Note: If this result is proved without Jordan forms, it can then be used as the basis of an alternative proof that every square matrix has a Jordan form.

- 114.** If V is a finite dimensional inner product space over $\mathbb{C}$ and T is a normal transformation on V , show that $\text{im}(T) \cap \ker(T) = \{\mathbf{0}\}$.

Hence show that every generalised eigenvector of T is actually an eigenvector, thereby showing that T is diagonalisable.

8 Functions of Matrices & ODEs

- 115.** Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$. Calculate $AB, BA, A^2 + 2AB + B^2, (A+B)^2, \exp(A)\exp(B), \exp(A+B)$ and $\exp(B)\exp(A)$. Which of these expressions might you have thought would be equal? Explain why they are not equal.

- [K]116.** Let A be a fixed $n \times n$ real matrix. Show that the set $G = \{\exp(tA)\}$ is a group of matrices, and deduce that G is a subgroup of $GL(n, \mathbb{R})$.

- [K]117.** Let $A = \begin{pmatrix} 7 & 8 \\ -6 & -7 \end{pmatrix}$. Show that A is diagonalisable. Hence find an expression for A^n for all $n \geq 0$ and use it to calculate $\exp(tA)$.

118. Let A be a square matrix which satisfies $A^3 = A^2$. Show that

$$\exp(A) = I + A + (e - 2)A^2. \text{ Hence find } \exp \begin{pmatrix} 0 & 2 & -1 \\ 1 & 2 & -2 \\ 0 & 2 & -1 \end{pmatrix}.$$

- [K]119.** (a) If A has eigenvalue λ , show that $\exp(A)$ has eigenvalue e^λ .
 (b) Show that $\exp(\text{tr}(A)) = \det(\exp(A))$. Deduce that $\exp(A)$ has positive determinant whenever A is real.
 (c) Find a B such that $\det(B) > 0$ and there is no real A with $\exp(A) = B$.

120. Show that

- (a) If A is normal, so is $\exp(A)$.
 (b) If A is skew-symmetric then $\exp(A)$ is orthogonal.
 (c) If A is skew-Hermitian ($A^* = -A$) then $\exp A$ is unitary.
 (d) There are orthogonal matrices that are *not* the exponential of any skew-symmetric matrix.
 (e) *Hard*: Prove that every unitary matrix is the exponential of some skew-Hermitian matrix. (In fact infinitely many, as those who have done MATH2621 or MATH2521 should be able to prove.)

[K]121. Find $\exp(tA)$ for each of the following diagonalisable matrices.

$$(a) \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}, \quad (b) \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}, \quad (c) \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

[K]122. Find $\exp(tA)$ for each of the following non-diagonalisable matrices A .

$$(a) \begin{pmatrix} 1 & 1 \\ -9 & -5 \end{pmatrix}, \quad (b) \begin{pmatrix} -1 & 1 & -1 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{pmatrix}, \quad (c) \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix}.$$

123. Let A be the matrix $\begin{pmatrix} 2 & 4 & 3 \\ 4 & 3 & 4 \\ -3 & -4 & -4 \end{pmatrix}$. Find a basis for $\mathbb{R}^3$ consisting of generalised eigenvectors for A , and use it to find e^{tA} .

124. Let $A = \frac{\pi}{2\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}$. Calculate the kernel of A , show that $Q = \exp(A)$ is a rotation and calculate the axis and angle of rotation.

[K]125. For each of the following matrices A , find $\exp(tA)$ by an appropriate method, and as a weak check show that your answer becomes the identity matrix when you put $t = 0$:

$$(a) \begin{pmatrix} 1 & 1 \\ -9 & -5 \end{pmatrix}, \quad (b) \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix}, \quad (c) \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad (d) \begin{pmatrix} -1 & 1 & -1 \\ 0 & 1 & -4 \\ 0 & 1 & -3 \end{pmatrix}.$$

[K]126. For each of the following matrices A , find the general solution to the system $\mathbf{y}' = A\mathbf{y}$ and then find the solution satisfying the initial condition $\mathbf{y}(0) = (1, 3)^T$.

$$(a) \begin{pmatrix} 1 & 2 \\ -1 & 4 \end{pmatrix}, \quad (b) \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}.$$

[K]127. For each of the following matrices A , find the general solution to the system $\mathbf{y}' = A\mathbf{y}$ and then find the solution satisfying the initial condition $\mathbf{y}(0) = (1, 2, 2)^T$.

$$(a) \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad (b) \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 1 \\ 0 & 1 & 3 \end{pmatrix}, \quad (c) \begin{pmatrix} 3 & -1 & 1 \\ 2 & 5 & -8 \\ 1 & 4 & -7 \end{pmatrix}.$$

128. Solve the system:

$$\begin{aligned} y_1' &= 3y_1 - y_2 + y_3, \\ y_2' &= 2y_1 + y_3, \\ y_3' &= 2y_1 - y_2 + 2y_3. \end{aligned}$$

with initial conditions $y_1(0) = y_3(0) = 1$, $y_2(0) = 0$.

129. Consider the second-order linear system $x'' = 3x - 2y$, $y'' = 2x - y$ with initial conditions $x(0) = 1$, $x'(0) = y(0) = y'(0) = 0$. Convert this system into a system of four first-order linear differential equations, and solve them for x and y . (Check your solution!!)

[K]130. Let

$$A = \begin{pmatrix} 8 & -1 & 4 \\ -4 & 5 & -4 \\ -7 & 2 & -3 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}.$$

- Calculate $(A - 3I)\mathbf{v}$ and $(A - 3I)^2\mathbf{v}$.
- Find bases for all generalised eigenspaces of A .
- Write $\mathbf{c}$ as a linear combination of these basis vectors.
- Solve the initial value problem $\mathbf{y}' = A\mathbf{y}$ with $\mathbf{y}(0) = \mathbf{c}$.

131. Let $A = \begin{pmatrix} 2 & 6 \\ 2 & 3 \end{pmatrix}$.

- Find the general solution to the system $\mathbf{y}' = A\mathbf{y}$.
- Check that if $\mathbf{b}(t) = \begin{pmatrix} -2t+7 \\ -2t+3 \end{pmatrix}$ then $\mathbf{y}(t) = \begin{pmatrix} t \\ -1 \end{pmatrix}$ is a solution to the non-homogeneous system $\mathbf{y}' = A\mathbf{y} + \mathbf{b}$. Write down the general solution to this system.
- For $\mathbf{b}(t) = \begin{pmatrix} 3e^{2t} \\ 2e^{2t} \end{pmatrix}$, find the solution to $\mathbf{y}' = A\mathbf{y} + \mathbf{b}$ with initial condition $\mathbf{y}(0) = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$.

132. In each case, find the solution to $\mathbf{y}' = A\mathbf{y} + \mathbf{b}$ with initial condition $\mathbf{y}(0) = \mathbf{c}$.

(a) $A = \begin{pmatrix} 3 & 6 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{b}(t) = \begin{pmatrix} -6e^{5t} \\ 3e^{5t} \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$

(b) $A = \begin{pmatrix} 11 & 9 & -14 \\ 2 & 4 & -2 \\ 12 & 12 & -15 \end{pmatrix}, \quad \mathbf{b}(t) = \begin{pmatrix} 6e^t \\ -2e^t \\ 3e^t \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$

[K]133. Solve the system $y_1'(t) = -3y_1 + y_2 + 3e^{-2t}$, $y_2'(t) = -y_1 - y_2 + 2e^{-2t}$, $y_1(0) = 1$, $y_2(0) = 1$.

[K]134. Let $A = \begin{pmatrix} 4 & 1 & -1 \\ 2 & 3 & -1 \\ 2 & 2 & 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

(a) Calculate $(A - 3I)\mathbf{v}$ and $(A - 3I)^2\mathbf{v}$.

(b) Show that the eigenvalues of A are 2 and 3.

(c) Find $\exp(tA)\mathbf{v}$ (without finding $\exp(tA)$).

(d) Solve the system $\mathbf{y}' = A\mathbf{y} + (e^{2t}, 0, 2e^{2t})^T$ with initial conditions $\mathbf{y}(0) = (1, 1, 1)^T$.

[K]135. Let $A = \begin{pmatrix} -2 & 0 & 1 & -2 \\ -2 & 1 & 0 & -2 \\ -3 & 0 & 2 & -2 \\ 1 & 0 & -1 & 1 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} 2 \\ 1 \\ 3 \\ -2 \end{pmatrix}.$

(a) Show that $\mathbf{u}$ is an eigenvector of A .

(b) Show that $\mathbf{v}$ is a generalised eigenvector for the eigenvalue 1 of A .

(c) Solve $\mathbf{y}' = A\mathbf{y} + e^{-t}\mathbf{u}$, $\mathbf{y}(0) = \mathbf{u} + \mathbf{v}$. (Do not blindly follow the basic algorithm but try to reduce the work involved.)

136. (a) Find the general solution of the system $\mathbf{y}'(t) = A\mathbf{y}(t)$ where A is the matrix

$$\begin{pmatrix} 11 & 9 & -14 \\ 2 & 4 & -2 \\ 12 & 12 & -15 \end{pmatrix}.$$

(b) Check that the non-homogeneous system $\mathbf{y}'(t) = A\mathbf{y} + \mathbf{b}(t)$ where

$$\mathbf{b}(t) = \begin{pmatrix} e^{2t} \\ -e^{2t} \\ 0 \end{pmatrix} \text{ has a solution } \mathbf{y}(t) = \begin{pmatrix} te^{2t} \\ -te^{2t} \\ 0 \end{pmatrix}. \text{ Write down the general solution.}$$

(c) Find the solution to $\mathbf{y}' = A\mathbf{y}(t) + \begin{pmatrix} 6e^t \\ -2e^t \\ 3e^t \end{pmatrix}$, given that $\mathbf{y}(0) = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$.

Selected Answers

1. (a) $\mathbf{x} = (2, 3, -2)$; Point of intersection of 3 planes.
 (b) $\mathbf{x} = (2, 3, -2, 0) + \lambda(-1, 0, 2, 1)$, $\lambda \in \mathbb{R}$; A line in $\mathbb{R}^4$ through the point $(2, 3, -2, 0)$ and parallel to $(-1, 0, 2, 1)$
2. (a) $\lambda = \pm 2$, (b) $\lambda = 1$, (c) all other values of λ .
3. (a) $AB = \begin{pmatrix} -11 & 7 \\ 2 & 1 \\ -11 & 8 \end{pmatrix}$. (b) BA is not defined. (c) $CD = \begin{pmatrix} -4 & -13 & -9 \\ -2 & 11 & 13 \\ 14 & 14 & 0 \end{pmatrix}$.
 (d) $DC = \begin{pmatrix} 3 & -6 \\ -23 & 4 \end{pmatrix}$ (e) $A^2 = \begin{pmatrix} -4 & -11 & 14 \\ 1 & -1 & 2 \\ -4 & -13 & 17 \end{pmatrix}$ (f) C^2 not defined.
 (g) $A^{-1} = \begin{pmatrix} -2 & -1 & 2 \\ 11 & 7 & -10 \\ 8 & 5 & -7 \end{pmatrix}$ (h) B^{-1} not defined. (i) $(CD)^{-1}$ does not exist.
 (j) $(DC)^{-1} = \frac{1}{126} \begin{pmatrix} -4 & -6 \\ -23 & -3 \end{pmatrix}$
6. $\mathbf{a}^T \mathbf{b} = \mathbf{b}^T \mathbf{a} = 8$, $\mathbf{ab}^T = \begin{pmatrix} 0 & 4 & 2 \\ 0 & 12 & 6 \\ 0 & -8 & -4 \end{pmatrix}$, $\mathbf{ba}^T = \begin{pmatrix} 0 & 0 & 0 \\ 4 & 12 & -8 \\ 2 & 6 & -4 \end{pmatrix}$, $\mathbf{ab}$ and $\mathbf{a}^T \mathbf{b}^T$ are not defined.
8. 1, -1, 0, 5, -2 respectively; all are invertible except (c).
9. -9, 0, 56 respectively.
19. (a) All elements are rotations. 6 have order 4, 8 have order 3 and 9 have order 2.
 (b) All elements are rotations, 24 have order 5, 20 have order 3 and 15 order 2.
21. (b) and (d) only 22. T, T, F, T 23. (b) and (d) only
25. T, F, F, T 27. (a) Yes. (b) No. (c) Yes. (d) No.
28. (a) No. (b) No. (c) Yes. (d) Yes. 29. (a) Yes. (b) No.
31. (a) No, for example, $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$. (b) No, for example, $\mathbf{v}_3 = 3\mathbf{v}_1 - 2i\mathbf{v}_2$.
33. yes, no, yes, yes 35. Coordinates are $(2, 0, 2, 1)^T$.
36. They are independent if $p \neq 5$. 37. All primes except $p = 2$ or 3.
38. Coordinates are $(4, -2, 1)^T$.
40. $W_1 + W_2 = V$, $\dim(W_1 \cap W_2) = 2$, so sum is not direct.

44. (a) linear, $\begin{pmatrix} 2 & 0 \\ 1 & -1 \end{pmatrix}$. (b) not linear. (c) linear, $\begin{pmatrix} 1 & 0 & 1 \\ 0 & -1 & -1 \\ 1 & 1 & 1 \end{pmatrix}$.

(d) linear, $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$.

45. (a) $(5, 7)^T$. (b) $T(\mathbf{x}) = (3x_1 - x_2, 3x_1 - 2x_2)^T$.

47. $\ker(A) = \text{span}(\{(5, -2, 1, 0, 0)^T, (5, -1, 0, 1, 0)^T, (0, 1, 0, 0, 1)^T\})$,
 $\text{col}(A) = \text{span}(\{(1, 1, 2, 3)^T, (3, 4, 3, 8)^T\})$,
 $\text{row}(A) = \text{span}(\{(1, 3, 1, -2, -3)^T, (1, 4, 3, -1, -4)^T\})$,
 $\text{nullity}(A) = 3$ and $\text{rank}(A) = 2$.

48. $\begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$. 49. $\begin{pmatrix} 10 & -9 \\ 5 & -5 \end{pmatrix}$.

51. $A = \begin{pmatrix} 15 & 10 & 17 \\ 4 & 3 & 5 \end{pmatrix}$ and $A' = \begin{pmatrix} 13 & 29 & 16 \\ 13 & 20 & 12 \end{pmatrix}$. 52. (c) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 6 \\ 0 & -2 & -3 \end{pmatrix}$

53. Neither

54. (a) $6 - 5i, 6 + 5i$. (b) $\frac{1}{3}(1 + 2i, 2)^T$.

55. (a) is an inner product; (b) is not positive definite;
 (c) is not symmetric or positive definite

56. (a) is an inner product; (b) is neither conjugate symmetric nor positive definite.

58. $\sqrt{22}$.

60. (a) $\{\frac{1}{13}(5, 12)^T, \frac{1}{13}(-12, 5)^T\}$. (b) $\{\frac{1}{3}(-2, 1, -2)^T, \frac{1}{3\sqrt{5}}(5, 2, -4)^T\}$.
 (c) $\{\frac{1}{7}(2, -3, 6)^T, \frac{1}{7\sqrt{2}}(9, 4, -1)^T\}$. (d) $\{\frac{1}{\sqrt{3}}(1, i, 1)^T, \frac{1}{2\sqrt{6}}(2i, 1 - 3i, 3 - i)^T\}$.

63. (a) $W^\perp = \text{sp}\{(1, 0, -1)^T\}$. (b) $(-4, 1, 5, 2)^T$.

64. (a) $W = \text{sp}\{\frac{1}{\sqrt{2}}(0, 1, 1, 0)^T, \frac{1}{2\sqrt{3}}(3, -1, 1, 1)^T\}$, $W^\perp = \text{sp}\{\frac{1}{2}(1, 1, -1, -1)^T, \frac{1}{\sqrt{6}}(0, 1, -1, 2)^T\}$.
 (b) $\mathbf{w}_1 = \frac{1}{4}(9, -1, 5, 3)^T$, $\mathbf{w}_2 = \frac{1}{4}(-5, -3, 3, 9)^T$.

65. One possibility is $\{\frac{1}{2}(1 + i, 1, -1)^T\}$.

66. (a) $T^* = T$, (b) $T^*(w, z) := (2w + z, -iw)$, (c) $T^*(v_1, v_2) = (2v_1 + \frac{1}{2}v_2, 2v_1)$.

70. (a) $\frac{1}{13} \begin{pmatrix} 5 & -12 \\ 12 & 5 \end{pmatrix} \begin{pmatrix} 13 & 4 \\ 0 & 6 \end{pmatrix}$. (b) $\frac{1}{3\sqrt{5}} \begin{pmatrix} -2\sqrt{5} & 5 \\ \sqrt{5} & 2 \\ -2\sqrt{5} & -4 \end{pmatrix} \begin{pmatrix} 3 & 6 \\ 0 & 3\sqrt{5} \end{pmatrix}$.
 (c) $\frac{1}{6} \begin{pmatrix} 3\sqrt{2} & -2\sqrt{3} \\ 0 & 2\sqrt{3} \\ 3\sqrt{2} & 2\sqrt{3} \end{pmatrix} \begin{pmatrix} \sqrt{2} & 3\sqrt{2} \\ 0 & \sqrt{3} \end{pmatrix}$. (d) $\frac{1}{6} \begin{pmatrix} 3 & \sqrt{6} & -3\sqrt{2} \\ 3 & -2\sqrt{6} & 0 \\ 3 & \sqrt{6} & 3\sqrt{2} \\ 3 & 0 & 0 \end{pmatrix} \begin{pmatrix} 2 & 2 & 2 \\ 0 & \sqrt{6} & -\sqrt{6}/2 \\ 0 & 0 & 3\sqrt{2}/2 \end{pmatrix}$.

- 71.** $2y = 3x - 2$. **72.** (a) $7y = 3x$. (b) $y = 1$. (c) $10y = 3x + 3$.
- 73.** (a) $2y = 5 - t$. (b) $4y = 3 - 9t + 7t^2$. **74.** $z_2 = (-2 - i) + (1 - i)z_1$
- 76.** a, b, c or $-a - b - c$. **77.** $(a - b)(b - c)(c - a)(a + b + c)$
- 84.** A : spectrum is $\{4, 5\}$. $E_4(A) = \text{span}\{(2, -3)^T\}$; $E_5(A) = \text{span}\{(2, -1)^T\}$; A is diagonalisable over $\mathbb{R}$, and so over $\mathbb{C}$.
 B : spectrum is $\{2, i, -i\}$ and $E_2(B) = \text{span}\{(1, 0, 0)^T\}$, so B is not diagonalisable over $\mathbb{R}$ but is over $\mathbb{C}$; $E_i(B) = \text{span}\{(1+i, 2, 3+i)^T\}$, $E_{-i}(B) = \text{span}\{(1-i, 2, 3-i)^T\}$.
 C : spectrum is $\{1, 2, 2\}$. $E_1(C) = \text{span}\{(0, -1, 1)^T\}$, $E_2 = \text{span}\{(0, 1, 0)^T\}$, so C is not diagonalisable over either $\mathbb{R}$ or $\mathbb{C}$.
- 85.** D : spectrum is $\{3+i, -1-i\}$ $E_{3+i}(D) = \text{span}\{(1, 1)^T\}$, $E_{-1-i}(D) = \text{span}\{(-1, 1)^T\}$, so D is diagonalisable over $\mathbb{C}$.
 F : spectrum is $\{i, i\}$, $E_i(F) = \text{span}\{(2, -1)^T\}$, so F is not diagonalisable over $\mathbb{C}$.
 G : spectrum is $\{1, 1+i, 1-i\}$, $E_1(G) = \text{span}\{(0, 0, 1)^T\}$,
 $E_{1+i}(G) = \text{span}\{(1, -1-i, 2)^T\}$, $E_{1-i}(G) = \text{span}\{(0, 1, 0)^T\}$, so G is diagonalisable over $\mathbb{C}$.
- 86.** (a) spectrum is $\{2, -3, 4\}$; $E_2(A) = \text{span}\{(1, 2, -2)^T\}$, $E_{-3}(A) = \text{span}\{(1, 0, 1)^T\}$; $E_4(A) = \text{span}\{(0, -1, 1)^T\}$. All algebraic multiplicities and geometric multiplicities are 1.
(b) $\text{span}\{(1, 0, 1)^T\} \subseteq E_3(B)$ and $\text{span}\{(2, -1, 2)^T\} \subseteq E_1(B)$. So spectrum is $\{3, 1, 4\}$. $E_4(B) = \text{span}\{(2, 2, -1)^T\}$. All algebraic multiplicities and geometric multiplicities are 1.
(c) spectrum is $\{5, 5, 5\}$. $E_5(C) = \text{span}\{(-1, 0, 1)^T, (2, 1, 0)^T\}$. 5 has algebraic multiplicity 3 and geometric multiplicity 2.
(d) $\text{span}\{(-2, 0, 0, 1)^T\} \subseteq E_3(D)$. But $\text{rank}(D - 3I) = 1$, so spectrum is $\{3, 3, 3, 0\}$. $E_3(D) = \text{span}\{(-2, 0, 0, 1)^T, (1, 0, 1, 0)^T, (-1, 1, 0, 0)^T\}$,
 $E_0(D) = \ker(D) = \text{span}\{(1, 2, 4, -1)^T\}$. 3 has algebraic multiplicity and geometric multiplicity 3; 0 has algebraic multiplicity and geometric multiplicity of 1
- 87.** (a) Yes; (b) No; (c) No; (d) Yes; (e) No; (f) Yes
- 88.** spectrum is $\{1, 1, 1, 1\}$, $E_1(T) = \text{constant polynomials}$.
- 94.** $\{\frac{1}{3}(1, -2, 2)^T, \frac{1}{3}(-2, 1, 2)^T, \frac{1}{3}(2, 2, 1)^T\}$
- 95.** **A**, $\lambda = 1, 3$, $E_1 = \text{sp}\{\frac{1}{\sqrt{2}}(1, -i)^T\}$, $E_3 = \text{sp}\{\frac{1}{\sqrt{2}}(1, i)^T\}$;
B, $\lambda = 3, -3$, $E_3 = \text{sp}\{\frac{1}{\sqrt{6}}(2 - i, 1)^T\}$, $E_{-3} = \text{sp}\{\frac{1}{\sqrt{6}}(1, -2 - i)^T\}$;
C, $\lambda = 0, 1, 3$, $E_0 = \text{sp}\{\frac{1}{\sqrt{3}}(1, 1, 1)^T\}$, $E_1 = \text{sp}\{\frac{1}{\sqrt{2}}(1, 0, -1)^T\}$, $E_3 = \text{sp}\{\frac{1}{\sqrt{6}}(1, -2, 1)^T\}$;
D, $\lambda = 1, 1, -1$, $E_1 = \text{sp}\{(0, 0, 1)^T, \frac{1}{\sqrt{2}}(1, 1, 0)^T\}$, $E_{-1} = \text{sp}\{\frac{1}{\sqrt{2}}(1, -1, 0)^T\}$.
- 96.** (a) $\{(1, 1)^T, (1, -1)^T\}$, hyperbola. (b) $\{(1, 1)^T, (1, -1)^T\}$, ellipse.
(c) $\{(2, -1)^T, (1, 2)^T\}$, 2 lines.
- 97.** (a) $\{(-2, 2, 1)^T, (1, 2, -2)^T, (2, 1, 2)^T\}$, ellipsoid. (b) $\{(1, 1, -2)^T, (1, 1, 1)^T, (-1, 1, 0)^T\}$, elliptical cylinder. (c) $\{(1, 1, 2)^T, (1, 1, -1)^T, (-1, 1, 0)^T\}$, hyperboloid of 2 sheets.
(d) $\{(-1, 0, 1)^T, (1, -2, 1)^T, (1, 1, 1)^T\}$, hyperboloid of 1 sheet.

- 98.** P is a rotation about $(-2, -2, 1)^T$ through π . Q is a rotation about $(\sqrt{2}, 0, -1)^T$ through $2\pi/3$. R is a reflection in the plane $x + y - 3z = 0$ composed with a rotation about $(1, 1, -3)^T$ through $\cos^{-1}(5/6)$.
- 99.** Singular values: for A are $\{5\sqrt{2}, 0\}$; for B are $\{\sqrt{7}, \sqrt{2}\}$; for C are $\{\sqrt{6}, 1, 0\}$
- 100.** $A \sim J_2(7)$, $B \sim J_2(-1)$ **101.** $A \sim J_3(3)$, $B \sim J_3(5)$, $C \sim J_2(2) \oplus (2)$
- 102.** $A \sim J_2(-3) \oplus (-5)$, $B \sim J_2(2) \oplus (-3)$, $C \sim (2) \oplus (-7) \oplus (-7)$
- 103.** $A \sim (5) \oplus (5) \oplus J_2(6)$. **104.** Yes, they have the same Jordan form $J_1(-1) \oplus J_2(1)$.
- 105.** $A \sim C \sim D$ **106.** There are 11 possible forms.
- 107.** (a) not possible as matrix is only 12×12 ; (b) $A \sim J_4(\lambda) \oplus J_3(\lambda) \oplus J_2(\lambda) \oplus J_2(\lambda) \oplus (\lambda)$, $\text{mp}_A(t) = (t - \lambda)^4$; (c) not possible as d_k cannot decrease; (d) $A \sim J_5(\lambda) \oplus J_4(\lambda) \oplus J_2(\lambda) \oplus (\lambda)$, $\text{mp}_A(t) = (t - \lambda)^5$; (e) and (f) not possible as $d_k - d_{k-1}$ cannot increase; (g) $A \sim J_5(\lambda) \oplus J_4(\lambda) \oplus J_3(\lambda)$, $\text{mp}_A(t) = (t - \lambda)^5$.
- 108.** (a) options are $J_6(3) \oplus J_2(3) \oplus (3)$, $J_5(3) \oplus J_3(3) \oplus (3)$, $J_4(3) \oplus J_4(3) \oplus (3)$; check nullities of $(A - 3I)^4$ and $(A - 3I)^5$; (b) options are $J_6(-2) \oplus J_3(-3) \oplus J_3(-2) \oplus J_2(-2)$, $J_5(-2) \oplus J_4(-3) \oplus J_3(-2) \oplus J_2(-2)$, $J_4(-2) \oplus J_4(-3) \oplus J_4(-2) \oplus J_2(-2)$, nullity $(A + 2I)^4$ can tell them apart; (c) options are $J_2(7) \oplus J_2(7) \oplus (7) \oplus (7) \oplus J_4(-3) \oplus J_2(-3)$ and $J_3(7) \oplus J_2(7) \oplus (7) \oplus (7) \oplus J_3(-3) \oplus J_2(-3)$; nullity $(A - 7I)^3$ will tell them apart.
- 110.** If either if A or B are invertible, minimal polynomials are the same. If both are singular, then the degrees of the min polys differ by at most 1 (they can differ by 1).
- 111.** $A^3 = 5A^2 - 8A + 4I$, $A^4 = 17A^2 - 36A + 20I$, $A^5 = 49A^2 - 116A + 68I$,
 $A^{-1} = \frac{1}{4}(A^2 - 5A + 8I)$.
- 113.** *Hint:* QR factorisation of the transition matrix.
- 115.** $AB = B$, $BA = A$, $A^2 + 2AB + B^2 = \begin{pmatrix} 4 & -2 \\ 0 & 0 \end{pmatrix}$, $(A + B)^2 = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}$
 $(\exp A)(\exp B) = \begin{pmatrix} e^2 & -(e - 1)^2 \\ 0 & 1 \end{pmatrix}$, $\exp(A + B) = \begin{pmatrix} e^2 & 0 \\ 0 & 1 \end{pmatrix}$,
 $(\exp B)(\exp A) = \begin{pmatrix} e^2 & (e - 1)^2 \\ 0 & 1 \end{pmatrix}$.
- 117.** $\exp(tA) = \cosh t I + \sinh t A$.
- 118.** $\begin{pmatrix} 2e - 3 & 2e - 2 & 5 - 3e \\ 2e - 3 & 2e - 1 & 4 - 3e \\ 2e - 4 & 2e - 2 & 6 - 3e \end{pmatrix}$.
- 121.** (a) $e^{2t} \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} + e^{3t} \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$, (b) $\frac{e^t}{5} \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} + \frac{e^{6t}}{5} \begin{pmatrix} 1 & -2 \\ -2 & 4 \end{pmatrix}$,
(c) $e^t \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$.

$$122. \text{ (a) } e^{-2t} \begin{pmatrix} 1+3t & t \\ -9t & 1-3t \end{pmatrix}, \quad \text{(b) } e^{-t} \begin{pmatrix} 1 & t+t^2/2 & -t-t^2 \\ 0 & 2t+1 & -4t \\ 0 & t & 1-2t \end{pmatrix},$$

$$\text{(c) } e^t \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 0 \end{pmatrix} + e^{2t} \begin{pmatrix} t+1 & -t & t \\ t+1 & -t & t \\ 1 & -1 & 1 \end{pmatrix}.$$

$$123. \begin{pmatrix} e^{3t} - te^{-t} & e^{3t} - e^{-t} & e^{3t} - (t+1)e^{-t} \\ e^{3t} - e^{-t} & e^{3t} & e^{3t} - e^{-t} \\ (t+1)e^{-t} - e^{3t} & e^{-t} - e^{3t} & (t+2)e^{-t} - e^{3t} \end{pmatrix}.$$

124. $\ker(A) = \text{span}\{(1, 0, 1)^T\}$. This is also the axis. The angle is $\pi/2$.

$$125. \text{ (a) } e^{-2t} \begin{pmatrix} 1+3t & t \\ -9t & 1-3t \end{pmatrix}; \quad \text{(b) } e^t \begin{pmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 0 \end{pmatrix} + e^{2t} \begin{pmatrix} t+1 & -t & t \\ t+1 & -t & t \\ 1 & -1 & 1 \end{pmatrix};$$

$$\text{(c) } e^{-t} \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}; \quad \text{(d) } e^{-t} \begin{pmatrix} 1 & t+t^2/2 & -t-t^2 \\ 0 & 2t+1 & -4t \\ 0 & t & 1-2t \end{pmatrix}.$$

$$126. \text{ (a) } \lambda e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \mu e^{2t} \begin{pmatrix} 2 \\ 1 \end{pmatrix} \text{ for } \lambda, \mu \in \mathbb{R}; \quad 5e^{3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} - 2e^{2t} \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

$$\text{(b) } \lambda e^t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \mu e^{6t} \begin{pmatrix} -1 \\ 2 \end{pmatrix} \text{ for } \lambda, \mu \in \mathbb{R}; \quad e^t \begin{pmatrix} 2 \\ 1 \end{pmatrix} + e^{6t} \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

$$127. \text{ (a) } \lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + \mu e^t \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} + \nu e^{3t} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \text{ for } \lambda, \mu, \nu \in \mathbb{R}; \quad \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix} + e^{3t} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

$$\text{(b) } e^{2t} \left(\lambda \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right) + \nu e^{4t} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \text{ for } \lambda, \mu, \nu \in \mathbb{R}; \quad e^{2t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2e^{4t} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

$$\text{(c) } \lambda e^{-3t} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + e^{2t} \left(\mu \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \nu \begin{pmatrix} t+1 \\ 2t \\ t \end{pmatrix} \right); \quad e^{2t} \begin{pmatrix} t+1 \\ 2t \\ t \end{pmatrix} + e^{-3t} \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$

$$128. \mathbf{y}(t) = \frac{-e^t}{2} \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \frac{3e^{3t}}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \quad 129. x = t \sinh t + \cosh t, y = t \sinh t.$$

$$130. \text{ (d) } \mathbf{y} = e^{3t}(-1 - 6t, 1 + 6t, 9t)^T + e^{4t}(-1, 0, 1)^T$$

$$131. \text{ (a) } \alpha e^{6t}(3, 2)^T + \beta e^{-t}(2, -1)^T. \quad \text{(b) } \alpha e^{6t}(3, 2)^T + \beta e^{-t}(2, 1)^T + (t, -1)^T.$$

$$\text{(c) } e^{-t}(-2, 1)^T + e^{6t}(15/4, 5/2)^T - e^{2t}(3/4, 1/2)^T.$$

$$132. \text{ (a) } \mathbf{y} = (0, e^{5t})^T. \quad \text{(b) } \mathbf{y} = e^{-3t}(1, 0, 1)^T + e^{2t}(1, -1, 0)^T + te^t(6, -2, 3)^T.$$

$$133. y_1 = (1 + 3t - t^2/2)e^{-2t}, y_2 = (1 + 2t - t^2/2)e^{-2t}.$$

$$134. \text{ (a) } (1, 1, 2)^T, (0, 0, 0)^T. \quad \text{(c) } e^{3t}(1 + t, 1 + t, 1 + 2t)^T.$$

$$\text{(d) } e^{3t}(1 + t, 1 + t, 1 + 2t)^T + e^{2t}(t, 0, 2t)^T.$$

135. (c) $\mathbf{y} = e^{-t}(1+t)\mathbf{u} + e^t(\mathbf{v} + t\mathbf{w})$, where $\mathbf{w} = (A - I)\mathbf{v} = (1, 0, 1, -1)^T$.

136. (a) $\mathbf{y}(t) = C_1 e^{-3t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + C_2 e^{2t} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + C_3 e^t \begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix};$

(c) $\mathbf{y}(t) = e^{-3t} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + e^{2t} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + t e^t \begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix}.$

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